

DARSHAN KUMAR G

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PROFESSIONAL SUMMARY

Instrumentation & Control Engineer currently working as a Graduate Engineer Trainee at Inox Air Products, with hands-on experience in the operation and process monitoring of a 200 TPD Liquid Air Separation Unit (ASU) and cryogenic processes. Skilled in instrument calibration, process analyzers, gauges, control valves, sensors, troubleshooting, and P&ID interpretation and development using AutoCAD 2D and AutoCAD Plant 3D. Experienced with Yokogawa CENTUM VP R6 DCS, including trip and interlock logic programming, control-loop tuning, trend monitoring, safe shutdown and cold startup procedures. Proficient in root cause analysis (RCA), fault diagnosis, process control, and instrumentation maintenance, with a strong focus on plant safety, reliability, and operational efficiency.

EDUCATION

B.E. - Electronics & Instrumentation Engineering

Dr Ambedkar Institute of Technology, Bengaluru, CGPA:**8.03**

(Dec 2023 – Aug 2026)

Diploma – Electrical & Electronics Engineering

PVP Polytechnic, Bengaluru, CGPA: **8.12**

(Dec 2020 – June 2023)

WORK EXPERIENCE

Graduate Engineer Trainee – Instrumentation

INOX Air Products Pvt. Ltd., Hosur, Tamil Nadu

(July 2026 – Present)

- Monitor and control a **200 TPD** Liquid Air Separation Unit (ASU), covering cryogenic process parameters, instrumentation, alarms, and plant operating conditions.
- Work with Yokogawa **CENTUM VP R6 DCS** for process monitoring, trend analysis, control-loop monitoring, and trip/interlock logic programming.
- Perform instrument calibration, analyzer monitoring, gauge inspection, troubleshooting, and maintenance of field instrumentation.
- Designed and developed P&IDs using **AutoCAD 2D**, **AutoCAD Plant 3D** as per company's MOC.
- Execute and support **safe shutdown** and **cold startup** operations while maintaining process safety and plant reliability.
- Perform control-valve tuning and control-loop optimization for stable and smooth process operation.
- Conduct **Root Cause Analysis** (RCA) and troubleshoot instrumentation, DCS, and process-related faults.
- Interpret **P&IDs**, **CLDs** and instrumentation documentation for plant tracing, troubleshooting, and process understanding.
- Monitor critical pressure, temperature, flow, level, and analyzer parameters to ensure safe, reliable, and efficient plant operation.

Application Engineer Intern

Digital Instruments & Control Systems, Bengaluru, India

(Jan 2026 – Apr 2026)

- Developed and tested PLC programs using **Allen-Bradley Studio 5000**, **Siemens TIA Portal V14**, Delta, Omron, Fuji, and Crouzet PLC platforms.
- Developed HMI applications using FactoryTalk View and Fuji HMI, including process visualization, control, alarms, and operator interfaces.
- Worked with industrial communication protocols including **Modbus RTU** (RS-232/RS-485), Modbus TCP/IP, **Ethernet**, **HART**, PROFIBUS-DP, and PROFIBUS-PA.
- Designed electrical schematics, control-panel layouts, and wiring diagrams using **AutoCAD Electrical**, **EPLAN Electric P8**, and **DesignSpark Electrical(Solid Works)**.
- Contributed to automation projects for **DRDO**, **Yokogawa**, and **Air Waters India**, gaining hands-on experience in PLC/HMI integration, industrial control systems, testing, debugging, and troubleshooting.

Electrical Design Intern

Innovation Innova CNC Pvt. Ltd, Bengaluru, India

(Mar 2023 – June 2023)

- Assisted in electrical **control-panel design**, schematic development, component selection, and electrical wiring for industrial automation systems.
- Designed and developed electrical circuits and wiring diagrams for **PLC, HMI, VFD**, and motor-control applications.
- Supported the design and assembly of VFD-based control panels for industrial applications, including projects for **Mahindra** and **3M India**.
- Worked with **AutoCAD Electrical** and electrical design documentation, gaining practical experience in panel layouts, wiring, and control-circuit design.
- Gained hands-on exposure to industrial electrical design, panel assembly, PLC/HMI systems, VFDs, commissioning, and troubleshooting.

TECHNICAL SKILLS

- **Instrumentation & Process Control:** Process Instrumentation, Sensors & Transmitters, Calibration, Control Valves, Process Analyzers, P&ID/PFD Interpretation, Control Loop Tuning, Troubleshooting, RCA
- **DCS & PLC Automation:** Yokogawa CENTUM VP R6, Trip & Interlock Logic, Safe Shutdown & Cold Startup, Allen-Bradley Studio 5000 / RS Logix 500, Siemens TIA Portal V14/V15, Delta, Omron, Fuji, Crouzet, HMI – FactoryTalk View & Fuji
- **Industrial Communication:** Modbus RTU (RS-232/RS-485), Modbus TCP/IP, Ethernet, HART, PROFIBUS-DP & PROFIBUS-PA
- **Electrical Design:** AutoCAD Electrical, AutoCAD Plant 3D, EPLAN Electric P8, DesignSpark Electrical, Electrical Schematics, Control Panel Design, Wiring Diagrams, SLDs, Panel Layouts, Component Selection
- **Standards:** IEC 61131-3 (PLC Programming), IEC 61511 (SIS), IEC 61158 (Industrial Communication Technology), IEC 60079 (Hazardous-area instrumentation)
- **Industrial Systems:** PLC/HMI/DCS, VFDs, Control Panels, Motors, Relays, Contactors, Sensors, Transmitters, Actuators, Cryogenic ASU Process Equipment
- **Industrial Safety:** EHS, LOTO (Lockout/Tagout), HAZOP, Permit to Work (PTW), Safety Work Permit (SWP), SOPs, Risk Assessment, Safe Shutdown & Startup, Process Interlocks, Trip Systems, Basic Process Safety

PROJECTS

- Automated Emergency Shower and Eyewash system
- Smart Street Lighting System using Piezoelectric Sensors & IoT Technology
- Arduino-Based Footstep Power Generation using Piezoelectric Transducers ([GitHub](#))
- Automatic Water Level Indicator ([GitHub](#))

SOFT SKILLS

- Lead a team of 4-members while developing the industrial projects during Internship.
- Detail-Oriented, Adaptability, Time Management, Attention to detail.
- Analytical Problem Solving, Troubleshooting & RCA, Technical Communication, Team Collaboration & Safety

CERTIFICATIONS

- **Electrical Design Certificate** – Innovation Innova CNC Pvt Ltd
- **Application Engineer Certificate**- Digital Instruments & Control Systems
- **Embedded System using Arduino** – GeeksforGeeks
- **Internet of Things** – Infosys | Springboard